Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization

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I. Poster Title

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Obstacle Avoidance & Occlusion-Aware UAV Trajectory Optimization.

II. Poster Authors

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III. Poster Abstract

Trajectory optimization for UAV target tracking in cluttered environments must simultaneously satisfy obstacle avoidance and occlusion-free visibility constraints, making the problem challenging for both control and optimization. This poster presents a discrete Higher-Order Unconstrained Binary Optimization (HUBO) formulation for obstacle-avoidance and occlusion-aware UAV trajectory optimization, designed to support quantum-classical benchmarking with quantum annealing methods and, after quadratization, QAOA. We also introduce a feasibility-aware evaluation methodology that separates hard-constraint satisfaction from soft trajectory quality, verifies decoded trajectories independently of energy thresholds, matches sampling budgets across stochastic solvers, and exposes penalty-weight exploitation through systematic sweeps. On an 8 x 8 benchmark instance with 960 variables, 87,684 monomials, and 3,172 cubic terms, we compare a classical A* baseline, simulated annealing (SA), D-Wave neal, and Qiskit QAOA. A* and SA achieve comparable soft-cost values of 317 and 303, respectively, while A* solves the instance in 10 ms and SA requires 47.8 s. Neal is much faster than SA at 0.26 s per sample, but reaches only 22% true feasibility; its feasibility rises from 0% at $\lambda = 50$ to 60% at $\lambda = 2000$. On a reduced 8-variable instance, QAOA reproduces the optimum and validates the formulation but does not scale to the full problem. The work clarifies when higher-order binary optimization may be useful for visibility-constrained UAV planning.

IV. Poster Relevance

This poster is directly relevant to QCE26 topics in quantum computing, quantum algorithms, quantum applications, and quantum software/systems. The work studies a hybrid quantum-classical optimization benchmark and evaluates quantum annealing and QAOA-style methods against classical baselines. It therefore aligns with topics including hybrid quantum-classical systems, adiabatic quantum computation and quantum annealing, variational quantum algorithms, QAOA, optimization techniques, algorithmic benchmarking, and resource estimation. The UAV trajectory problem also connects to quantum applications in optimization, transportation, logistics, and decision making, while the feasibility-aware evaluation protocol supports quantum systems software topics such as simulators, development kits, tools, workflows, and benchmarking. By emphasizing decoded-trajectory feasibility rather than raw energy alone, the poster addresses a practical validation issue for near-term quantum optimization research and engineering practice.

Submission structure: This PDF contains Sections 1–4 on this page, Section 5 as the two-page extended poster abstract on the following two pages, and Section 6 as the one-page preliminary poster on the final page.

Feasibility-Aware Quantum-Classical Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Trajectory Optimization

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***Abstract*—Trajectory optimization for UAV target tracking in cluttered environments must simultaneously satisfy obstacle-avoidance and occlusion-free visibility constraints, making the problem highly challenging for both control and optimization. This paper presents a discrete Higher-Order Unconstrained Binary Optimization (HUBO) formulation for occlusion-aware UAV trajectory optimization, designed to enable quantum-classical benchmarking with quantum annealing methods and, after quadratization, QAOA. In addition, we introduce a feasibility-aware evaluation methodology that separates hard-constraint satisfaction from soft trajectory quality, applies independent constraint verification, matches sampling budgets across stochastic solvers, and characterizes penalty-weight exploitation through systematic sweeps. On an 8×8 benchmark instance comprising 960 variables, 87,684 monomials, and 3,172 cubic terms, we compare a classical A* baseline, simulated annealing (SA), D-Wave neal, and Qiskit QAOA. A* and SA achieve comparable soft-cost values (317 and 303, respectively), while A* solves the problem in 10 ms and SA requires 47.8 s. Neal is substantially faster than SA (0.26 s per sample) but attains only 22 % true feasibility, with feasibility increasing from 0 % at $\lambda{=}50$ to 60 % at $\lambda{=}2000$. On a reduced instance, QAOA reproduces the optimum and serves as a correctness check, but does not scale to the full problem size. The inclusion of a classical occlusion-aware baseline grounds the HUBO formulation against a conventional planner and clarifies the regime in which higher-order binary optimization may become useful.**

***Index Terms*—HUBO, quantum annealing, QAOA, occlusion-aware tracking, UAV trajectory, A*, feasibility, benchmarking.**

I. Introduction

Visibility-aware UAV target tracking — keeping a moving target in view while avoiding obstacles — has an established literature in robotics and control. Chen *et al.* [1] plan collision-free quadrotor trajectories tracking a target in cluttered environments. Occlusion and perception have been treated explicitly: Hausman *et al.* [2] address occlusion-aware multi-robot tracking, Penin *et al.* [3] combine target tracking with collision and occlusion avoidance, Falanga *et al.* [4] fold perception objectives into MPC, and Wang *et al.* [5] extend to multi-UAV visibility-constrained tracking. Masnavi *et al.* [6] introduced the occlusion-free target-tracking and obstacle-avoidance problem solved here as a real-time multi-convex MPC. All of these are *continuous* control formulations.

In the quantum-optimization setting, QUAV [7] addresses single-UAV obstacle-avoidance path planning with QAOA on small instances, and Q4DR [8] solves drone *routing* via a hybrid QAOA-clustering + D-Wave decomposition, closer to logistics than to local motion. Neither addresses visibility-constrained trajectory planning.

The contribution of this paper is not a claim of quantum speedup, but the introduction of a quantum-compatible HUBO benchmark for occlusion-aware and obstacle-avoidance UAV trajectory optimization, and a feasibility-aware methodology for evaluating such models. The inclusion of classical baselines is deliberate: current quantum hardware limitations make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment establish a clear path toward future QUBO/QAOA and D-Wave hardware studies.

Concretely, our contribution comprises **(1)** a discrete HUBO reformulation of the occlusion-aware target-tracking class [6], bridging continuous MPC and higher-order binary optimization; **(2)** a feasibility-aware benchmarking methodology with a classical A* baseline; and **(3)** an analysis of why the sustained-proximity term requires auxiliary variables for quadratization.

II. HUBO Reformulation

Let V be the grid cells and T the horizon. The decision $x[t,v]\in\{0,1\}$ is 1 iff the UAV occupies cell v at time t. The Hamiltonian has seven terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \quad (1)$$

H_{uniq}, H_{start}, H_{move} enforce kinematic constraints quadratically; H_{obs} forbids obstacle cells.

Occlusion penalty. For each cell u we trace the Bresenham line-of-sight to the target and count intervening obstacles:

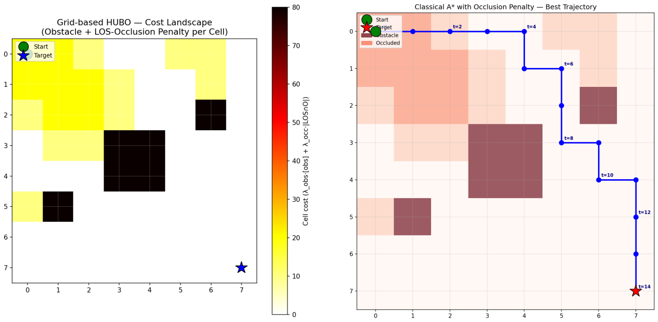
$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t} \sum_{u\in V} x[t,u] \,\big|\, \text{LOS}(u,\text{target}) \cap O \,\big|, \quad (2)$$

with O the obstacle set (Fig. 1a). For a static target this is linear; for a *moving* target it becomes a joint product of UAV and target occupancy — intrinsically higher-order.

Genuinely-cubic proximity. The sustained-proximity term penalizes lingering in a buffer cell v_b for three consecutive steps:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b\in B} \sum_{t=1}^{T-1} x[t{-}1,v_b]\, x[t,v_b]\, x[t{+}1,v_b]. \quad (3)$$

Eq. (3) requires auxiliary variables for quadratization: Rosenberg reduction introduces one ancilla per cubic term [9],



(a) Occlusion cost surface. **(b)** A* path: 80 % visibility.

Fig. 1: (a) Per-cell obstacle + line-of-sight occlusion cost to the target; yellow cells are occluded. (b) A* trajectory: feasible, target-reaching.

adding ∼3,172 ancillas (∼30 % more variables) with dense couplings that degrade the sparsity annealers exploit. The full instance has **960 variables**, **87,684 monomials** (3,172 cubic).

III. Feasibility-Aware Benchmarking

Raw HUBO energy misleads because penalty-based solvers can lower energy by *violating* soft constraints. Our methodology adopts four safeguards: **(i)** energy decomposition into hard penalty and soft cost; **(ii)** independent constraint verification on each decoded trajectory (not energy-threshold heuristics); **(iii)** matched sampling budgets across stochastic solvers; and **(iv)** a penalty-weight sweep revealing the exploitation regime a single λ would hide. We add a **classical A* baseline** to ground the HUBO against a conventional planner.

Since Qiskit's state-vector simulator scales as 2^n and cannot run at n=960, we use two tiers: **Tier 1** (full 8×8, 960 vars: A*, SA, neal) and **Tier 2** (tiny 2×2, 8 vars: all solvers including QAOA). We use 50 matched samples per stochastic solver, SA at 500 sweeps, and hardened weights λ=1000. A* runs on the same grid with cell cost $1.0 + \lambda_{\text{occ}}|\text{LOS} \cap O| + \lambda_{\text{prox}}\mathbf{1}[v_b \in B]$ and Manhattan heuristic; it is an *approximate analogue* of the HUBO surrogate, encoding goal-reaching through search termination rather than H_{term}.

IV. Results

A. Tier 1 — Three Solvers, One Instance

Table I reports the matched comparison. A* and SA achieve comparable soft cost (317 vs. 303), but A* completes in 10 ms while SA needs 47.8 s — a ∼4,700× speedup with guaranteed feasibility. Neal is fast per sample (0.26 s) but only 22 % of its samples are truly feasible, and its best feasible soft cost (496) is 56 % worse than A*'s (Fig. 1b). SA's *mean* soft cost (∼495) matches neal's *best* (496). Visibility is reported for the A* path; the stochastic benchmark logged per-sample energy and feasibility but not full traces, so an SA/neal visibility score would require re-running with trajectory logging.

B. Constraint-Exploitation Curve

Sweeping λ (Fig. 2) reveals the failure mode the methodology targets. At $\lambda \leq 300$ neal achieves 0 % feasibility; it rises to 13 % at λ=600, 20 % at λ=1000, and 60 % at λ=2000. A single-λ report would mis-state neal's capability.

TABLE I: Tier 1 head-to-head, 8×8 instance (960 variables).

Metric	A*	SA	neal
Best soft cost	317	**303**	496
Feasibility rate	**100 %**	100 %	22 %
Reaches target	**yes**	14 % runs	8 % runs
Visibility (% steps)	**80 %**	—	—
Runtime per sample	**10 ms**	47.8 s	0.26 s

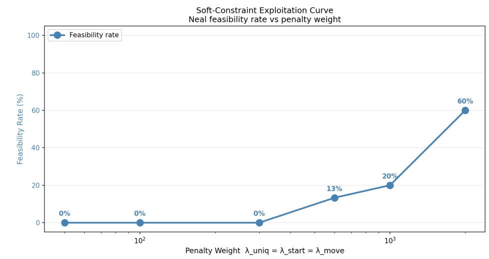


Fig. 2: Constraint-exploitation curve: neal feasibility vs. penalty weight λ. Annealers need weights an order of magnitude above the objective scale to respect one-hot HUBO structure.

C. Tier 2 — Three-Way Comparison with QAOA

On the 8-variable instance, A*, SA, neal, and Qiskit QAOA all reach the same optimum $E^* = -118.17$ (QuadraticProgram → QAOA, SPSA, reps=1), included for *correctness validation*, not performance competition. QAOA takes 55 s vs. 0.02 s (neal) and 0.14 s (SA): a 2,750× state-vector overhead that grows exponentially with qubit count.

V. Discussion and Conclusion

The A* baseline answers the obvious reviewer question: the HUBO captures sensible planning structure — A* finds a feasible trajectory in milliseconds with soft cost within 5 % of the best stochastic solver. The framing's value is not raw performance on this static-target instance but compatibility with quantum-style solvers and extensibility to a *moving* target, where Eq. (2) becomes intrinsically cubic and outside A*'s shortest-path framing. Feasibility-aware benchmarking proves necessary: 78 % of neal's hardened-weight samples violate constraints. Future work: moving target, real D-Wave hardware, and comparison with the multi-convex MPC of [6].

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SCOPE

We introduce a quantum-compatible HUBO benchmarkfor occlusion-aware UAV trajectory optimization and a **feasibility-aware methodology** for evaluating such models. Classical baselines are deliberate: current quantum hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment chart a path toward future D-Wave / QAOA hardware studies.

1. The Problem

UAV target tracking in cluttered space must **simultaneously**:

- avoid obstacles, and
- keep **line-of-sight** to a target (occlusion-free).

Masnavi *et al.* solved this as continuous multi-convex MPC. We recast the **same problem class** as a discrete higher-order binary problem for quantum solvers.

2. HUBO Formulation

Binary $x[t, v] = 1$ iff UAV at cell v, time t. Seven cost terms:

$$H = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}$$

Occlusion (Bresenham line-of-sight):

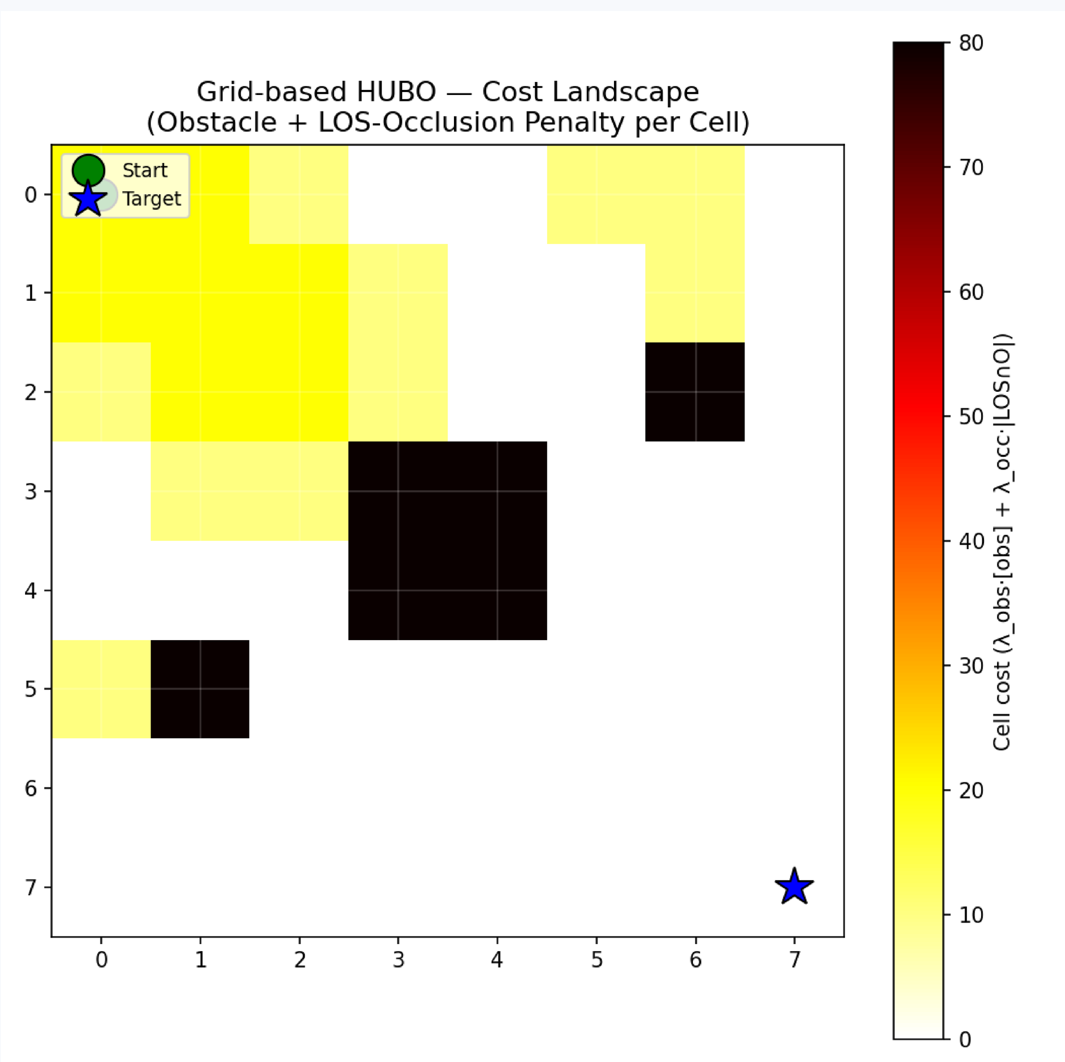
$$H_{\text{occ}} = \lambda_{\text{occ}} \textstyle\sum_t \sum_u x[t, u]\, |\text{LOS}(u, \text{tgt}) \cap O|$$

Genuinely cubic proximity (lingering in a buffer cell):

$$H_{\text{prox}} = \lambda_{\text{prox}} \textstyle\sum_{v_b} \sum_t x[t{-}1, v_b]\, x[t, v_b]\, x[t{+}1, v_b]$$

Why HUBO not QUBO? Quadratizing the cubic term needs $\sim$3,172 ancillas ($\sim$30% more variables) with dense couplings that wreck sparsity.

Occlusion Cost Surface



Yellow = target occluded from that cell; dark = obstacle.

3. Feasibility-Aware Benchmarking

Raw HUBO energy **misleads** — annealers lower energy by *violating* soft constraints. Four safeguards:

(i) Energy decomposition
hard-penalty vs. soft-cost, kept separate

(ii) Independent constraint check
decode + verify, not energy thresholds

(iii) Matched sampling budgets
equal reads/runs across solvers

(iv) Penalty-weight sweep
exposes the constraint-exploitation regime

Two-Tier Design

State-vector QAOA scales as 2^n — impossible at n=960.

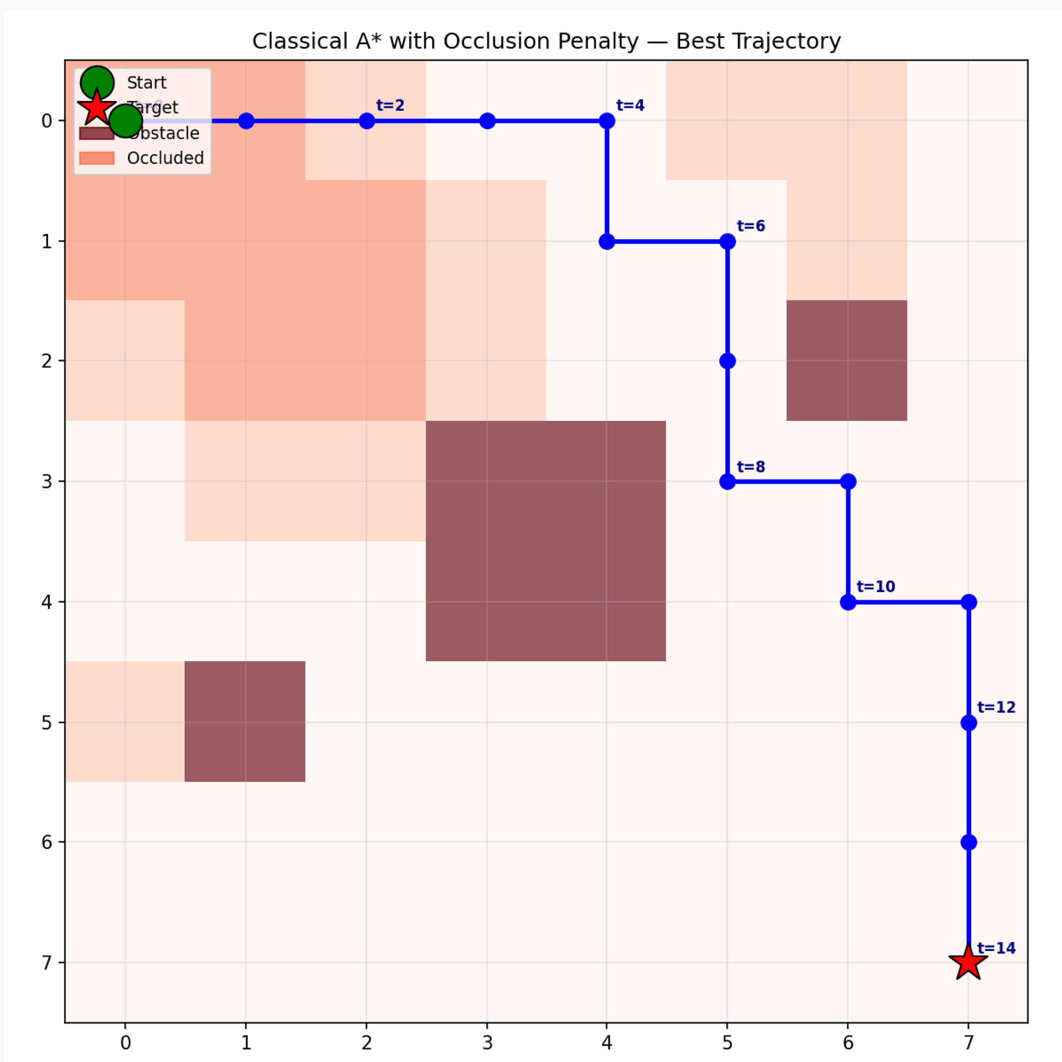
TIER 1	**TIER 2**
Full 8×8	Tiny 2×2
960 vars	8 vars
A* · SA · neal	+ Qiskit QAOA

Classical A* Baseline

A* on the same grid with occlusion-weighted cell cost

$$1.0 + \lambda_{\text{occ}} |\text{LOS} \cap O| + \lambda_{\text{prox}} \mathbf{1}[v_b \in B]$$

grounds the HUBO against a conventional planner — answering whether the formulation captures **sensible planning** or is a benchmark artifact.



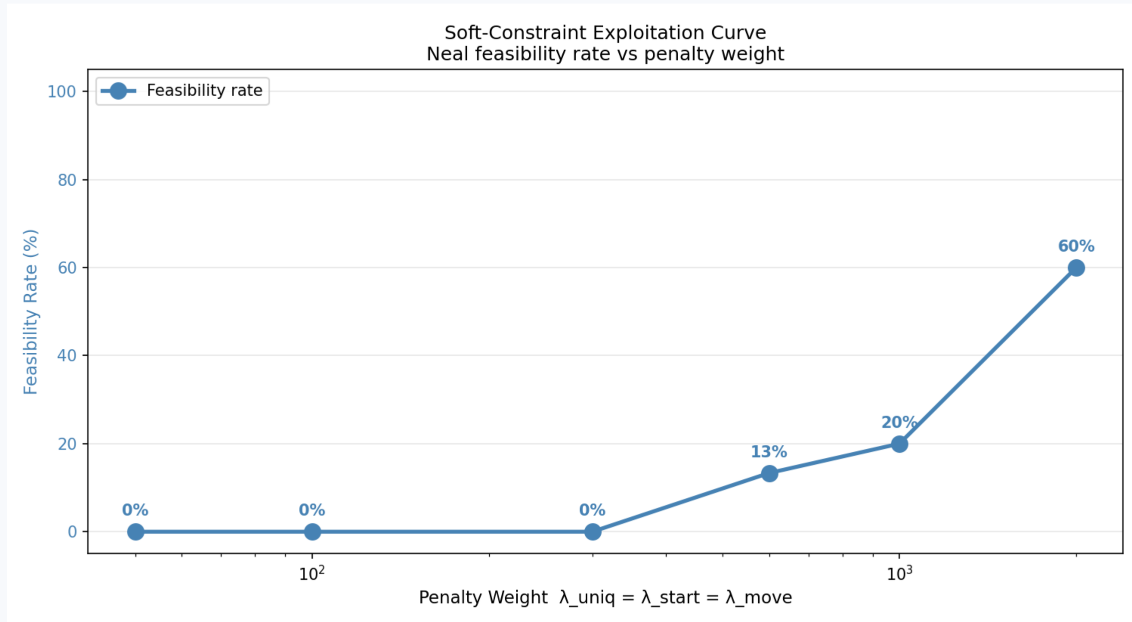
A* path: feasible, reaches target, 80% visibility.

4. Results: 8×8 Instance

Metric	A*	SA	neal
Soft cost	317	**303**	496
Feasibility	**100%**	100%	22%
Reaches target	**yes**	14%	8%
Visibility	**80%**	—	—
Runtime/sample	**10 ms**	47.8 s	0.26 s

A* and SA reach comparable quality; A* is $\sim$4,700× faster. neal is fast but only 22% feasible.

Headline: Constraint-Exploitation Curve



neal feasibility climbs **0%** → **60%** as λ: 50 → 2000

General-purpose annealers need penalty weights an **order of magnitude** above the objective scale to respect one-hot HUBO structure. A single λ would misreport feasibility.

Tier 2 & Takeaways

On 8 variables, **all four solvers** reach the same optimum ($E^* = -118.17$); QAOA is a correctness check (2,750× slower, no scaling).

The HUBO's value is not raw speed here, but compatibility with quantum solvers and extensibility to a **moving target**, where occlusion becomes intrinsically cubic — beyond A*'s shortest-path reach.

Future: moving target · real D-Wave hardware · vs. continuous MPC.

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